Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (1pt)**.** Let G be a bipartite graph with parts U, W, and fix an integer $k \geqslant 1$. Show that if $\deg u \geqslant k$ for all $u \in U$ and $\deg w \leqslant k$ for all $w \in W$, then G has a matching which saturates U.

Solution. This is a direct consequence of the last Theorem on Lecture 18, which states that G contains a matching which saturates U if $\deg u \geqslant \deg w$ for all $u \in U$, $w \in W$ such that $uv \in E(G)$. □

Problem 2 (2pts)**.** Let G be a graph on $n \geqslant 2$ vertices with $\delta(G) \geqslant n/2$. Show that:

(a) If n is even, then G has a perfect matching.

(b) If n is odd, then $G - v$ has a perfect matching for every $v \in V(G)$.

Hint: Hamiltonian cycles.

Solution. Consider first the case of $n = 2$; since $\delta(G) \geqslant 1$ here, we know that $G \cong K_2$, which has a perfect matching.

Now assume $n \geqslant 3$. Since $\partial(G) \geqslant n/2$, by Dirac's theorem we know that G contains a Hamiltonian cycle, say $(v_1, \ldots, v_n)$.

(a) Suppose that n is even. Then the edges $v_1v_2, v_3v_4, \ldots, v_{n-1}v_n$ are vertex-disjoint and cover all vertices of G. In other words, these edges form a perfect matching.

(b) Suppose that n is odd and fix any $v \in V(G)$. Since $(v_1, \ldots, v_n)$ is a Hamiltonian cycle of G, we may suppose, without loss of generality, that $v_n = v$. Then the edges $v_1v_2, v_3v_4, \ldots, v_{n-2}v_{n-1}$ are vertex-disjoint and cover all vertices of G except for $v_n = v$. In other words, these edges form a perfect matching for $G - v$. □

Problem 3 (2pts)**.** Let G be a k-regular bipartite graph. Prove that we can partition the edges of G into k perfect matchings. That is, show that we can can partition $E(G) = M_1 \sqcup \cdots \sqcup M_k$ where each M_i is a perfect matching in G.

Solution. We prove the claim by induction on k. For the base case $k = 0$, the graph G has no edges and so $E(G) = \varnothing$ can trivially be partitioned into $k = 0$ many perfect matchings.

Assume now $k \geqslant 1$. By Theorem 8.6, we know that G contains a perfect matching, which we denote M_k. Consider the graph $H = G - M_k$. Since M_k is a perfect matching for G, we know that every vertex of G is incident to exactly one edge in M_k, so H is $(k-1)$-regular. Moreover, H is bipartite because it is a spanning subgraph of G. Hence, by the induction hypothesis, we can partition $E(H) = M_1 \sqcup \cdots \sqcup M_{k-1}$ into where each M_i is a perfect matching for H. But, since H is a spanning subgraph of G, these M_i are also perfect matchings for G. Thus $E(G) = M_1 \sqcup \cdots \sqcup M_{k-1} \sqcup M_k$ gives the desired partition. □

Problem 4 (2pts)**.** Let G be a bipartite graph with parts U, W wherein no vertex of U is isolated. Show that if all vertices in U have distinct degrees, then G contains a matching which saturates U.

Solution. We verify Hall's condition, so fix any non-empty $X \subseteq A$; we must show that $|N(X)| \geqslant |X|$. Set $k = |X|$. We claim that there is some $x \in X$ with $\deg x \geqslant k$. Indeed, if $\deg x \leqslant k - 1$ for all $x \in X$, since no vertex of $X \subseteq U$ is isolated then $\deg x \in \{1, \ldots, k-1\}$ for all $x \in X$. But since $k - 1 < k = |X|$, this would force two vertices in X to have the same degree, a contradiction. Thus, since $N(X) \supseteq N(x)$ for each $x \in X$, we have $|N(X)| \geqslant \deg x \geqslant k = |X|$, which concludes the proof. □

Problem 5 (1+2pts)**.** Let G be an n-vertex graph. Recall that the **degeneracy** of a graph G is

$$d(G) = \max\{\delta(H) \colon H \text{ is a subgraph of } G\}$$

(a) Show that there is an ordering $V(G) = \{v_1, \dots, v_n\}$ such that $|N(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d(G)$ for all $i \in \{1, \dots, n\}$.

Hint: induct by deleting an appropriate vertex.

(b) Use the ordering on part (a) to give an alternative proof of Theorem 10.9, i.e. $\chi(G) \leqslant 1 + d(G)$.

Solution. (a) We proceed inductively on n. For the base case $n = 1$ we have $V(G) = \{v_1\}$, which certainly verifies the claim.

Assume now $n \geqslant 2$. Let v_n denote a vertex with $\deg v = \delta(G)$, and consider the $(n-1)$-vertex graph $H = G - v_n$. Note that, since H is a subgraph of G, we have $d(H) \leqslant d(G)$. By induction hypothesis, there is an ordering $V(H) = \{v_1, \dots, v_{n-1}\}$ such that $|N_H(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d(H)$ for all $i \in \{1, \dots, n-1\}$. Consider the ordering $V(G) = \{v_1, \dots, v_{n-1}, v_n\}$. We need to show that

$$|N_G(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d(G), \text{ for all } i \in \{1, \dots, n\}$$

Take first $i \in \{1, \dots, n-1\}$. Since $H = G - v_n$, we see that $v_n \notin \{v_1, \dots, v_{i-1}\}$, thus

$$|N_G(v_i) \cap \{v_1, \dots, v_{i-1}\}| = |N_H(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d(H) \leqslant d(G).$$

Lastly, for $i = n$, we have

$$|N_G(v_n) \cap \{v_1, \dots, v_{n-1}\}| \leqslant |N_G(v_n)| = \deg_G v_n = \delta(G) \leqslant d(G),$$

where the last inequality follows since G is a subgraph of itself.

(b) We define a proper coloring $f \colon V(G) \to \{1, \dots, d(G) + 1\}$ using the ordering from part (a).

Start by setting $f(v_1) = 1$. Next, fix $i \in \{2, \dots, n\}$. Assume recursively that f has already been defined on $\{v_1, \dots, v_{i-1}\}$, and define $f(v_i)$ to be the smallest color not used by the already-colored neighbors of v_i (that is, the smallest color not used in $N(v_i) \cap \{v_1, \dots, v_{i-1}\}$). Since there are $d(G)+1$ many available colors and $|N(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d(G)$, this is always possible.

The resulting coloring f is a proper coloring. Indeed, if $v_i v_j \in E(G)$ where $i < j$, then $v_i \in N(v_j) \cap \{v_1, \dots, v_{j-1}\}$, and so $f(v_i) \neq f(v_j)$ by construction.

Since f uses at most $d(G) + 1$ many colors, this shows that $\chi(G) \leqslant d(G) + 1$. □

Problem 6 (1 bonus pt)**.** Let G be an n-vertex graph. Show that $d(G)$ is the smallest integer d such that there is an ordering $V(G) = \{v_1, \dots, v_n\}$ so that $|N(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d$ for all $i \in \{1, \dots, n\}$.

This problem will be graded all-or-nothing.

Solution. We know that such an ordering exists for $d(G)$. Assume that d is some integer for which there is an ordering $V(G) = \{v_1, \dots, v_n\}$ so that $|N(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d$ for all $i \in \{1, \dots, n\}$. We need to show that $d \geqslant d(G)$.

To do so, consider any subgraph H of G. Let $i \in \{1, \dots, n\}$ be the smallest index for which H is a subgraph of the induced graph $G[\{v_1, \dots, v_i\}]$. Now we show that $\delta(H) \leqslant d$. Indeed, since H is a subgraph of $G[\{v_1, \dots, v_i\}]$, we have

$$N_H(v_i) = N_H(v_i) \cap \{v_1, \dots, v_{i-1}\} \subseteq N_G(v_i) \cap \{v_1, \dots, v_{i-1}\}.$$

Hence

$$\delta(H) \leqslant \deg_H(v_i) = |N_H(v_i)| \leqslant |N_G(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leqslant d$$

In other words, $\delta(H) \leqslant d$ for any subgraph H of G. Thus $d(G) \leqslant d$. □